

C++ LANGUAGE QUICK CHEATSHEET

LAST-MINUTE PLACEMENT EXAM REVISION (1-2 MIN SCAN)



CLASSES & OBJECTS

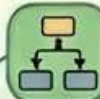
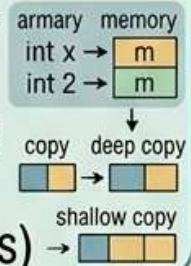
- class syntax
- constructors & destructors
- static members
- this pointer

```
class {
    access:
        access/er;
        access:optunt;
    specifiers:
        access:
            sthss pointer;
};
```



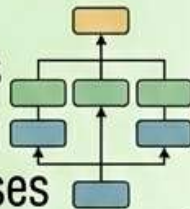
THE BIG THREE (Rule of 3)

- Copy Constructor
`*ptr = *var;`
- Copy Assignment Operator (=)
- Destructor (~Class)



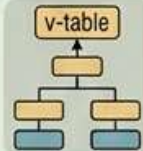
INHERITANCE

- single, multiple, multilevel ...
- access specifiers (public, protected)
- virtual base classes



POLYMORPHISM

- Compile-time (Overloading)
- Run-time (Overriding, Virtual Fn) (Overriding, Virtual Fn)
- Abstract Classes & Pures Virtual Fns



TEMPLATES & EXCEPTIONS

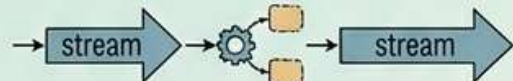
- Function Templates
- Class Templates
- Exception Handling (try-catch)

```
template <->      class
void *metane({, ...} anjec() {
    try () {
        ...
    } catch {}
```



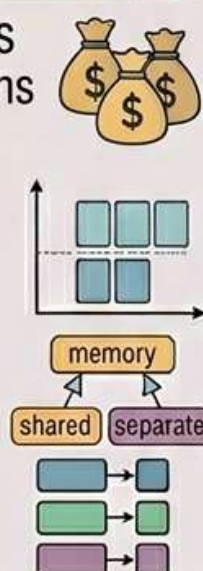
FILE I/O & PREPROCESSOR

- fstream, ifstream, ofstream
- using namespace std;
- preprocessor directives (#define, #include)



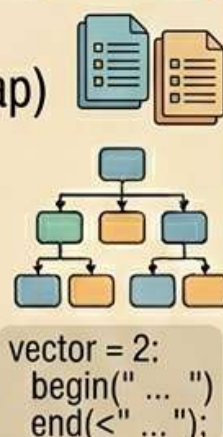
C++11 & MODERN FEATURES

- auto, lambda fns
 - auto, lambda fns
- smart pointers (unique_ptr, shared_ptr)
- nullptr
- initializer lists
 - ... = NC+
 - ... = Aft
 - ...



STL (Containers & Algorithms)

- Containers (vector, list, map)
- Iterators (begin, end)
- Algorithms (sort, find)



CONSOLIDATED REVISION ZONE

